S1 Practice Problems

30 problems across Algebra, Calculus 1, and Differential Equations.

Algebra (Problems 1–4)

1. Solve for x: $\dfrac{2x-3}{x+1} = \dfrac{x+5}{x-2}$

2. Find all real solutions: $x^4 - 5x^2 + 4 = 0$

3. Solve: $\log_2(x+3) + \log_2(x-1) = 5$

4. Perform partial fraction decomposition:

$$\frac{5x-1}{(x-1)(x+2)}$$

Limits (Problems 5–7)

5. Evaluate:

$$\lim_{x \to 0} \frac{\sin(3x)}{x}$$

6. Evaluate:

$$\lim_{x \to \infty} \frac{4x^3 - 2x + 1}{7x^3 + 5x^2}$$

7. Evaluate:

$$\lim_{x \to 0} \frac{e^{2x} - 1 - 2x}{x^2}$$

Basic Derivatives (Problems 8–10)

8. Differentiate: $f(x) = 4x^5 - \frac{3}{x^2} + \sqrt{x} - 7$

9. Differentiate: $f(x) = 3\sin(x) - 5\cos(x) + 2\tan(x)$

10. Differentiate: $f(x) = e^{3x} - \ln(x^2 + 1)$

Chain, Product, and Quotient Rules (Problems 11–13)

11. Find $\dfrac{dy}{dx}$: $\quad y = x^3 \ln(x) - e^{2x} \sin(x)$

12. Find $\dfrac{dy}{dx}$: $\quad y = \dfrac{x^2 - 1}{\sin(x)}$

13. Find $\dfrac{dy}{dx}$: $\quad y = \cos^3(4x^2 + 1)$

Implicit Differentiation (Problems 14–15)

14. Find $\dfrac{dy}{dx}$: $\quad x^2y + y^3 = 6$

15. Find $\dfrac{dy}{dx}$ and the equation of the tangent line at $\left(\dfrac{3}{2}, \dfrac{3}{2}\right)$: $\quad x^3 + y^3 = 3xy$

Applied Optimization (Problems 16–17)

16. Find the absolute maximum and minimum of $f(x) = x^3 - 3x^2 - 9x + 5$ on $[-2, 4]$.

17. A particle's position is $s(t) = t^3 - 6t^2 + 9t$ for $t \geq 0$. Find when the particle is at rest, when it is moving in the positive direction, and its total distance traveled on $[0, 4]$.

Basic Integrals (Problems 18–20)

18. Evaluate:

$$\int \left(6x^2 - \frac{4}{x} + 3e^x\right) dx$$

19. Evaluate:

$$\int_0^\pi \sin(x)\,dx$$

20. Use the Fundamental Theorem of Calculus to find $F'(x)$:

$$F(x) = \int_1^{x^2} \frac{t}{t^2+1}\,dt$$

U-Substitution (Problems 21–23)

21. Evaluate:

$$\int \frac{3x^2 + 2}{x^3 + 2x + 1} \, dx$$

22. Evaluate:

$$\int_0^1 x\, e^{x^2}\, dx$$

23. Evaluate:

$$\int \sin^4(x) \cos(x) \, dx$$

Integration by Parts (Problems 24–25)

24. Evaluate:

$$\int_0^{\pi/2} x\cos(x)\,dx$$

25. Evaluate:

$$\int x^2 e^x \, dx$$

Differential Equations (Problems 26–30)

26. Solve the separable ODE:

$$\frac{dy}{dx} = \frac{x^2}{y}, \quad y(0) = 3$$

27. Solve the first-order linear ODE:

$$\frac{dy}{dx} + 2y = 4e^{-x}$$

28. Find the general solution:

$$y'' - 5y' + 6y = 0$$

29. Solve the initial value problem:

$$y'' + 4y' + 4y = 0, \quad y(0) = 1,\ y'(0) = -1$$

30. A tank contains 100 L of brine with 10 kg of salt dissolved. Pure water flows in at 5 L/min and the well-mixed solution drains at the same rate. Set up and solve the IVP for the amount of salt $Q(t)$ at time t. How long until only 1 kg remains?